

Statistical Inference

Class Test - 2

28 October 2025

Total Marks: 10

Name: _____

Roll Number: _____

Instructions

- Answer as much as you can. The maximum you can score is 10.
- Only write the answer in the space provided or tick the correct option.
- Any additional writing will not be graded. Rough work should be done on spare pages.

Question 1. [1 mark]

Let $X_1, \dots, X_n \stackrel{i.i.d}{\sim} \text{Uniform}(0, \theta)$, $\theta > 0$. Define $T_1 = 2\bar{X}$ and $T_2 = \frac{n+1}{n}X_{(n)}$ to be two estimators of θ . Which of the following is true?

- (a) T_1 is unbiased, T_2 is biased.
- (b) T_2 is unbiased, T_1 is biased.
- (c) T_1 is the UMVUE of θ .
- (d) T_2 is the UMVUE of θ .

Question 2. [1 mark]

Let $X_1, \dots, X_n \stackrel{i.i.d}{\sim} \text{Normal}(\mu, 1)$, $\mu \in \mathbb{R}$. Let $\theta = \mu^2$ be the parameter of interest. Define $T = \bar{X}^2 - \frac{1}{n}$. Which of the following is correct?

- (a) T is biased for θ .
- (b) T is unbiased for θ , but not the UMVUE.
- (c) T is the UMVUE of θ , but it does not attain the Cramér-Rao lower bound.
- (d) T attains the Cramér-Rao lower bound.

Question 3. [1 mark]

Let $I_n(\theta)$ denote the Fisher information about θ based on a random sample of size n from the Poisson(θ) distribution with $\theta > 0$. Then, which of the following is correct?

- (a) $I_{2n}(\theta) = 4I_n(\theta)$
- (b) $I_{2n}(\theta) = 2I_n(\theta)$
- (c) $I_{2n}(\theta) = \frac{I_n(\theta)}{4}$

(d) $I_{2n}(\theta) = \frac{I_n(\theta)}{2}$

Question 4. [1.5 marks]

Let $X_1, \dots, X_n \stackrel{i.i.d}{\sim} \text{Normal}(\theta, \theta^2)$, $\theta > 0$. Consider $T = (\sum_{i=1}^n X_i, \sum_{i=1}^n X_i^2)$. Which of the following is correct?

- (a) T is sufficient for θ , but not complete.
- (b) T is sufficient for θ and complete.
- (c) T is not sufficient for θ , but complete.
- (d) T is neither sufficient for θ , nor complete.

Question 5. [1.5 marks]

Let $X_1, \dots, X_n \stackrel{i.i.d}{\sim} \text{Normal}(\mu, \sigma^2)$, $\mu \in \mathbb{R}, \sigma^2 > 0$, where both μ and σ^2 are unknown. Let $\theta = (\mu, \sigma^2)$ and $g(\theta) = \mathbb{P}_\theta(X_1 > 5)$. Define:

$$\bar{X} = \frac{1}{n} \sum_{i=1}^n X_i, \quad S^2 = \frac{1}{n} \sum_{i=1}^n (X_i - \bar{X})^2$$

Here, Φ is the cdf of $\text{Normal}(0, 1)$. Which of the following is the MLE of $g(\theta)$?

- (a) $T_1 = \Phi\left(\frac{5-\bar{X}}{S}\right)$
- (b) $T_2 = 1 - \Phi\left(\frac{5-\bar{X}}{S}\right)$
- (c) $T_3 = \Phi\left(\frac{5-\mu}{\sigma}\right)$
- (d) $T_4 = 1 - \Phi\left(\frac{5-\mu}{\sigma}\right)$
- (e) None

Question 6. [2 marks]

Let $X \sim \text{Poisson}(\theta)$ and $Y \sim \text{Poisson}(3\theta)$ be two independent random variables, where $\theta > 0$ is the unknown parameter. Find the MLE of θ based on (X, Y) .

Answer:

Question 7. [2 marks]

Let $X_1 \sim \text{Exponential}(\theta)$ and $X_2 \sim \text{Exponential}(2\theta)$ be independent, where $\theta > 0$. Find the Fisher information on θ contained in the sample (X_1, X_2) .

Hint: The pdf of Exponential(θ) distribution is $f_\theta(x) = \theta e^{-\theta x}, x \geq 0$.

Answer:

Question 8. [2 marks]

Let $X_1, \dots, X_n$ be i.i.d. with pdf:

$$f_\theta(x) = \begin{cases} \frac{2}{\theta} x e^{-\frac{x^2}{\theta}} & x \geq 0 \\ 0 & \text{otherwise} \end{cases}$$

where $\theta > 0$. Find the Fisher information on θ contained in the sample $(X_1, \dots, X_n)$.

Answer: